

Zhenyu Wu

Applied Scientist — LLMs, Structured Data & Knowledge Graphs

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Summary

Final-year Computer Science PhD candidate specialising in LLM-based structured-data understanding, schema and taxonomy inference, knowledge graphs, and agentic data analytics. Experienced in building reproducible research systems with PyTorch, Hugging Face Transformers, vLLM, and LangGraph.

Experience

University of Manchester , Doctoral Researcher — LLMs for Structured Data and Knowledge Construction	Manchester, UK 2022 – 2026 4 years 2 months
- Designed encoder- and decoder-based systems for inferring entity types, attributes, hierarchies, and semantic relationships from heterogeneous tabular repositories. - Improved F1 and Recall by 30–40% and Rand Index by 7–10% over prior approaches across real-world datasets. - Built modular Python pipelines with PyTorch, Hugging Face Transformers, and vLLM for batched inference, caching, structured validation, ablations, and reproducible evaluation. - Developing an agentic multi-database question-answering system combining schema linking, structured reasoning, Text-to-SQL, and tool calling with Qwen2.5-14B.	
Singapore University of Technology and Design , Visiting Research Associate	Singapore 2021 – 2022 1 year
- Co-developed a unified benchmark for data-stream clustering, published in Proceedings of the ACM on Management of Data (SIGMOD 2023). - Implemented CluStream, DenStream, DStream, and DBStream within a shared C++ codebase and reproducible evaluation framework.	

Education

PhD University of Manchester , Computer Science	Manchester, UK 2022 – 2026
- Expected completion October 2026 - Dean's Doctoral Scholarship	
MSc University of Edinburgh , High Performance Computing with Data Science	Edinburgh, UK 2019 – 2020
Graduated with Merit	
BSc Tianjin University of Finance and Economics , Information Management and Information Systems	Tianjin, China 2015 – 2019
- GPA: 3.81/4.00; Top 5% - Minor in Accounting	

Publications

Schema Inference for Tabular Data Repositories Using Large Language Models <i>Zhenyu Wu</i> , Jiaoyan Chen, Norman W. Paton arxiv.org/abs/2509.04632 (The VLDB Journal)	2026
Taxonomy Inference for Tabular Data Using Large Language Models <i>Zhenyu Wu</i> , Jiaoyan Chen, Norman W. Paton 10.1007/978-3-031-94575-5_22 (The Semantic Web — ESWC 2025)	2025
Dataset Discovery and Exploration: A Survey Norman W. Paton, Jiaoyan Chen, <i>Zhenyu Wu</i> 10.1145/3626521 (ACM Computing Surveys)	2024

Data Stream Clustering: An In-depth Empirical Study

2023

Xin Wang, Zhengru Wang, *Zhenyu Wu*, Shuhao Zhang, Xuanhua Shi, Li Lu
[10.1145/3589307](#) (Proceedings of the ACM on Management of Data)

Skills

Programming: Python, SQL, C++, Linux/Shell, Git
Machine Learning and LLMs: PyTorch, Hugging Face Transformers, vLLM, self-supervised learning, contrastive learning, hierarchical clustering
Structured Data and Knowledge: Schema and taxonomy inference, ontology construction, knowledge graphs, table understanding, entity typing, attribute resolution
Agents and Data Intelligence: LangChain, LangGraph, tool calling, structured-output validation, schema linking, Text-to-SQL

Awards

Dean’s Doctoral Scholarship University of Manchester
2022

Languages

Mandarin: Native
English: Professional proficiency